

1 Appendix 1: tables of constants and reference/scaling values used

Table 1: Constants used by Schubert et al., 2001

Constant	Value	Name
k_b	0	Bulk viscosity
α		thermal expansivity
χ_T		isothermal compressibility
c_v		specific heat at constant volume
c_p		specific heat at constant pressure
χ_a		adiabatic compressibility
k		thermal conductivity
H		rate of internal heat production per unit mass
K_a		adiabatic bulk modulus
K_T		isothermal bulk modulus

Table 2: Reference values used by Schubert et al., 2001

Symbol	Value	Description
ΔT		Characteristic temperature difference driving thermal convection
χ_{T_r}		Representative isothermal compressibility
α_r		Representative thermal expansivity
u_r	$\frac{k_r}{\rho_r c_{p_r} b}$	Velocity scale
μ_r	10^{21} Pa s	Representative viscosity
χ_a		Adiabatic compressibility
b		Depth of convecting region
p_r	$\frac{\mu_r u_r}{b}$	Scale for dynamic pressure variations
k_r	$4 \text{ W m}^{-1} \text{ K}^{-1}$	Reference value thermal conductivity
c_{p_r}	$1 \text{ kJ kg}^{-1} \text{ K}^{-1}$	Reference specific heat at constant pressure

Table 3: Dimensionless parameters and their values for the Earth’s mantle

Parameter	Definition	Value	Description
Pr	$\frac{\mu_r c_{p_r}}{k_r}$	2.5×10^{23}	Prandtl number
M ²	$\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}$	10^{-33}	Mach number (squared)
ϵ	$\alpha_r \Delta T_r$		Fractional density change due to temperature variations